SL No.X/Sem4 /I

Model Question Paper I

Energy Management and Auditing

Time: 3 Hours *Max.Marks: 75*

PART A

Answer **all** questions in one word or one sentence. Each question carries **1 mark**.

1	Define Energy audit	M1.03	R
2	Which is the instrument used for measuring illumination level	M1.04	R
3	Name any two types of Boilers	M2.02	R
4	List two properties of Refractories	M2.03	R
5	What is the other name of cogeneration	M2.04	R
6	Name any two types of motors	M3.02	R
7	List the compressed air system components	M3.03	R
8	List any one limitation of simple payback period	M4.02	R
9	List one advantage of NPV	M4.02	R

PART B

Answer any **eight** questions from the following, each question carries **3 mark**s.

(8 * 3=24 marks)

1	Explain the classification of Energy sources	M1.01	U
2	Explain the working principle of Lux meters	M1.04	U
3	What is the purpose of Insulation	M2.03	R
4	List any two energy saving measures in pump system	M3.04	R
5	List the advantages of an energy efficient motor	M3.02	R
6	Explain HVAC system	M3.03	U
7	List any two energy saving measures in cooling tower	M3.04	R
8	What is the Role of ESCO	M4.03	R
9	Explain cash flow model	M4.03	U
10	List three advantages of Average rate of return method	M4.02	R

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Describe Ultrasonic flow meters and Stroboscopic Non contact tachometers	M1.04	U
	OR		
IV	Explain the duties of Energy managers	M1.02	U
V	Explain Detailed energy audit steps	M1.03	U
	OR		
VI	Explain Benchmarking	M1.03	U
VII	Explain the types and classification of Furnaces	M2.02	U
	OR		
VIII	Explain energy conservation opportunities in Boilers	M2.02	U
IX	Explain the classification of Insulation	M2.03	U
	OR		
X	Illustrate back pressure steam turbine cogeneration systems	M2.04	U
XI	List any seven energy conservation opportunities in Lighting system	M3.01	R
	OR		
XII	List energy management opportunities in Electric motors	M3.02	R

<table>
<tr><td>XIII</td><td>Solve and estimate the payback period if the electricity charges are rupees 7 per Kwh for a non star Refrigerator to be placed with star refrigerator(220 liters). Assume following details

<table>
<tr><td></td><td>Star rated device</td><td>Non Star rated device</td></tr>
<tr><td>cost</td><td>Rs 34000/-</td><td>Rs 20000/-</td></tr>
<tr><td>Star level</td><td>5 star</td><td>No star</td></tr>
<tr><td>Annual Electricity consumption</td><td>816 units</td><td>2080 units</td></tr>
</table>
</td><td>M4.02</td><td>A</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>XIV</td><td>A new small cogeneration plant installation is expected to reduce a company's annual energy bill by Rs 486000. If the capital cost of the new boiler installation is Rs 2220000 and the annual maintenance and operating cost are 42000,calculate the expected payback period for the project.</td><td>M4.02</td><td>A</td></tr>
</table>

Module Wise Question Analysis

Module	Hr/ module	(hi / ∑Hi) * 123	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Marks	No	Marks	No	Marks	No	Marks
I	10	28.60	2	2	2	6	4	28	**8**	**36**
II	12	34.32	3	3	1	3	4	28	**8**	**34**
III	11	31.46	2	2	4	12	2	14	**8**	**28**
IV	10	28.60	2	2	3	9	2	14	**7**	**25**
Total			9	9	10	30	12	84	**31**	**123**

Cognitive Level Wise Question Analysis

level	% mark distribution	marks/level	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Marks	No	Marks	No	Marks	No	Marks
R	30	36.9	9		6		2		**17**	
				9		18		14		**41**
U	50	61.5	0		4		8		**12**	
				0		12		56		**68**
A	20	24.6	0		0		2		**2**	
				0		0		14		**14**
Total			9		10		12		**31**	
				9		30		84		**123**

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.03	R	1	2
I.2	M1.04	R	1	2
I.3	M2.02	R	1	2
I.4	M2.03	R	1	2
I.5	M2.04	R	1	2
I.6	M3.02	R	1	2
I.7	M3.03	R	1	2
I.8	M4.02	R	1	2
I.9	M4.02	R	1	2
II.1	M1.01	U	3	8
II.2	M1.04	U	3	8
II.3	M2.03	R	3	7
II.4	M3.04	R	3	7
II.5	M3.02	R	3	7
II.6	M3.03	U	3	8
II.7	M3.04	R	3	7

II.8	M4.03	R	3	7
II.9	M4.03	U	3	7
II.10	M4.02	R	3	7
III.	M1.04	U	7	17
IV.	M1.02	U	7	17
V	M1.03	U	7	17
VI	M1.03	U	7	17
VII	M2.02	U	7	17
VIII	M2.02	U	7	17
IX	M2.03	U	7	17
X	M2.04	U	7	17
XI	M3.01	R	7	17
XII	M3.02	R	7	17
XIII	M4.02	A	7	17
XIV	M4.02	A	7	17
Total			123	295

Prepared By :

Fathima Salim
Assistant Professor
EEE Dept.
College of Engineering Pathanapuram

Scrutinized By :

Sasna.N.A
Lecturer in Electrical and Electronics Engineering
Sreerama Govt. Polytechnic College, Triprayar

SL No.X/Sem4/II

Model Question Paper II

Energy Management and Auditing

Time: 3 Hours *Max.Marks: 75*

PART A

Answer **all** questions in one word or one sentence. Each question carries **1 mark**.

1	What is the role of energy manager	M1.02	R
2	Define Energy Management	M1.02	R
3	Name the 2 parts of a boiler	M2.02	R
4	Define cogeneration	M2.04	R
5	Name two commercial waste heat recovery devices	M2.04	R
6	Name the type of Lamp which is an energy efficient replacement for Fluorescent Lamp	M3.01	R
7	Define Coefficient of performance	M3.03	R
8	List two advantages of Internal rate of return method	M4.02	R
9	List two advantages of simple payback period	M4.02	R

PART B

Answer any **eight** questions from the following, each question carries **3 mark**s.

(8 * 3=24 marks)

1	List the Responsibilities of Energy manager	M1.02	R
2	What is the principle of combustion of fuels	M2.01	R
3	Explain any three Prime Movers for Cogeneration and give a brief outline about each	M2.04	U
4	Explain a Boiler system	M2.02	U
5	List various types of combustion controls	M2.01	R
6	What is the necessity of cogeneration	M2.04	R
7	List energy saving measures in Air Conditioning System	M3.04	R
8	Explain the ESCO concept	M4.03	U
9	List the Benefits of Energy monitoring and Targeting	M4.01	R
10	Explain need and benefits of star labeling	M4.04	U

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Explain Energy conservation and its importance	M1.01	U
	OR		
IV	Describe Radiation Pyrometers and Power Analyzers	M1.04	U
V	What is meant by targeted energy audit	M1.03	R
	OR		
VI	What is Preliminary Energy Audit Methodology	M1.03	R
VII	Explain energy conservation opportunities in Furnaces	M2.02	U
	OR		
VIII	Summarize the classification of Refractories	M2.03	U
IX	Explain any seven energy conservation opportunities in Compressed Air system	M3.03	U
	OR		
X	Outline any seven energy conservation opportunities in Refrigeration Plants	M3.03	U
XI	Explain any seven energy conservation opportunities in DG set	M3.04	U
	OR		
XII	Explain energy saving measures in electrolytic process	M3.02	U

<table>
<tr><td>XIII</td><td>Solve and estimate the payback period if the electricity charges are rupees 8 per Kwh for a non star Refrigerator to be placed with star refrigerator(220 liters). Assume following details

<table>
<tr><td></td><td>Star rated device</td><td>Non Star rated device</td></tr>
<tr><td>cost</td><td>Rs 32000/-</td><td>Rs 18000/-</td></tr>
<tr><td>Star level</td><td>5 star</td><td>No star</td></tr>
<tr><td>Annual Electricity consumption</td><td>800 units</td><td>2050 units</td></tr>
</table></td><td>M4.02</td><td>A</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>XIV</td><td>A cogeneration system installation is expected to reduce a company's annual energy bill by RS 23 Lakhs.If the capital cost of the new cogeneration installation is Rs 90 Lakhs and the annual maintenance and operating costs are Rs 5 lakhs,solve the expected pay back period for the project</td><td>M4.02</td><td>A</td></tr>
</table>

Module Wise Question Analysis

Module	Hr/ module	(hi / ΣHi) * 123	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Marks	No	Marks	No	Marks	No	Marks
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IV	10	28.60	2	2	3	9	2	14	**7**	**25**
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Cognitive Level Wise Question Analysis

level	% mark distribution	marks/level	Type of questions							
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				9		18		14		**41**
U	50	61.5	0		4		8		**12**	
				0		12		56		**68**
A	20	24.6	0		0		2		**2**	
				0		0		14		**14**
Total			9		10		12		**31**	
				9		30		84		**123**

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.02	R	1	2
I.2	M1.02	R	1	2
I.3	M2.02	R	1	2
I.4	M2.04	R	1	2
I.5	M2.04	R	1	2
I.6	M3.01	R	1	2
I.7	M3.03	R	1	2
I.8	M4.02	R	1	2
I.9	M4.02	R	1	2
II.1	M1.02	R	3	7
II.2	M2.01	R	3	7
II.3	M2.04	U	3	8
II.4	M2.02	U	3	8
II.5	M2.01	R	3	7
II.6	M2.04	R	3	7
II.7	M3.04	R	3	7
II.8	M4.03	U	3	8
II.9	M4.01	R	3	7
II.10	M4.04	U	3	7
III.	M1.01	U	7	17
IV.	M1.04	U	7	17
V	M1.03	R	7	17

VI	M1.03	R	7	17
VII	M2.02	U	7	17
VIII	M2.03	U	7	17
IX	M3.03	U	7	17
X	M3.03	U	7	17
XI	M3.04	U	7	17
XII	M3.02	U	7	17
XIII	M4.02	A	7	17
XIV	M4.02	A	7	17
Total			123	295

Prepared By :

Fathima Salim
Assistant Professor
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College of Engineering Pathanapuram

Scrutinized By :

SIGNATURE

Sasna.N.A
Lecturer in Electrical and Electronics Engineering
Sreerama Govt. Polytechnic College, Triprayar

Scoring Indicators

Model Question Paper I

Energy Management and Auditing

Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			9
I. 1	**Energy audit** *definition: 1 mark*	1	1	
	As per the Energy Conservation Act, 2001, Energy Audit is defined as "*the verification, monitoring and analysis of use of energy including submission of technical report containing recommendations for improving energy efficiency with cost benefit analysis and an action plan to reduce energy consumption*".			

I. 2	**Instrument used for measuring illumination level** *1 mark*	1	1	
	Lux meter			
I. 3	**Two types of boilers** *1 mark*	1	1	
	There are mainly two types of boiler – water tube boiler and fire tube boiler.			
I. 4	**Two properties of Refractories** *0.5mark each* *2* 0.5 marks = 1 mark*	0.5+ 0.5	1	
	Important properties of refractories include chemical composition, bulk density, apparent porosity, apparent specific gravity and strength at atmospheric temperature			
I. 5	**Other name of cogeneration** *1 mark*	1	1	
	Combined heat and power (CHP)			
I. 6	**Any two types of motors** *0.5mark each* *2* 0.5 marks = 1 mark*	0.5+0.5	1	
	1)Induction motor 2)DC motor			
I. 7	**Compressed air systems usually consist of the following components:** ***Any two,*** *0.5mark each* *2* 0.5 marks = 1 mark*	0.5+0 .5	1	

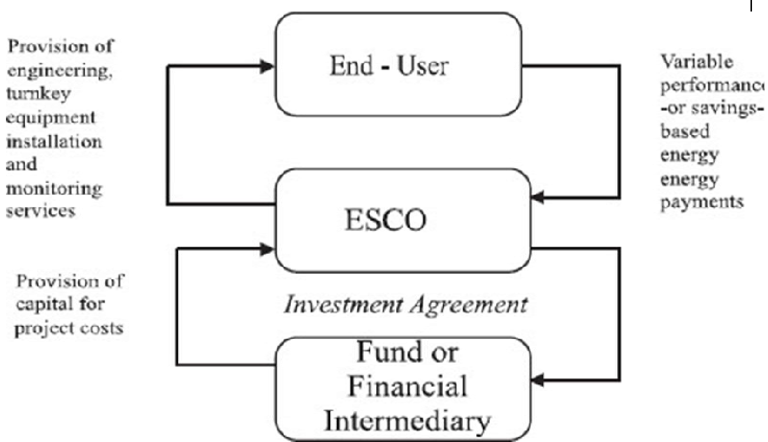
	• compressor. • air cooler. • air receiver tank. • filter. • dryer. • condensate trap. • distribution system.			
I. 8	**Limitation of simple payback period** *1 mark*	1	1	
	Ignores the time value of money:			
I. 9	**Advantage of NPV** *1 mark*	1	1	
	• The advantages of the net present value includes the fact that it considers the time value of money and helps the management of the company in the better decision making			
	PART B			24
II. 1	**classification of Energy sources** *classification-3 marks*	3	3	
	Energy can be classified into several types based on the following criteria: • Primary and Secondary energy • Commercial and Non commercial energy • Renewable and Non-Renewable energy			
II. 2	**Lux meters** *3 marks*	3	3	

	Most of the lux meters consist of a body, photocell or light sensor, and display. The light that falls on to the photocell or sensor contains energy that is converted to electric current. In turn, the amount of current depends on the light that strokes the photocell or light sensor. Lux meters read the electrical current, calculate the appropriate value, and show this value on its display. It is the basic lux meter working principle. Analog lux meter display values on a dial usually with a needle or pointer whereas digital devices display values as numbers letters. A Lux meter may be portable or bench type. Selecting lux meters or light meters requires certain performance specifications including photocell, illumination range, Lux resolution, operating temperature and foot candle resolution. Special features include low battery Indicators, low voltage, alarms, remote light sensors, built-in memory, auto power off, zero function etc.			
II. 3	**Purpose of Insulation** *3 mark*	3	3	

	A thermal insulator is a poor conductor of heat and has a low thermal conductivity. Insulation is used in buildings and in manufacturing processes to prevent heat loss or heat gain. Although its primary purpose is an economic one, it also provides more accurate control of process temperatures and protection of personnel. It prevents condensation on cold surfaces and the resulting corrosion. Such materials are porous, containing large number of dormant air cells. Thermal insulation delivers the following benefits: • Reduces over-all energy consumption • Offers better process control by maintaining process temperature. • Prevents corrosion by keeping the exposed surface of a refrigerated system above dew point • Provides fire protection to equipment • Absorbs vibration			
II. 4	**Any two energy saving measures in pump system** *Any two-1.5marks each*	3	3	

	. Ensure availability of basic instruments at pumps like pressure gauges, flow meters. Operate pumps near the best efficiency point. Modify the pumping system and pump losses to minimize throttling. Adapt to wide load variation with variable speed drives or sequenced control of multiple units. Stop running multiple pumps - add an auto-start for an on-line spare or add a booster pump in the problem area. Use booster pumps for small loads requiring higher pressures.			
II. 5	**Energy efficient motors advantages** *any six- 0.5 marks each*	0.5*6	3	
	1 The EEM has greater efficiency than a standard motor. 2 EEM has lower slip so they have a higher speed than standard motor 3 It can reduce maintenance cost and improve operations in industry due to robustness and reliability 4 Operate more satisfactory under abnormal voltage 5 Electric power saving 6 Operating temperature less 7 Noise level is lower			
II. 6	**HVAC system** *explanation- 3 marks*	3	3	

	HVAC stands for Heating, Ventilation, and Air Conditioning. HVAC systems control the indoor environment (temperature, humidity, air flow, and air filtering) Mechanical intervention to condition the air to a preferred temperature and relative humidity. HVAC is a basic requirement for our indoor air quality.The goal of the heating, ventilating, and air conditioning (HVAC) system is to create and maintain a comfortable environment within a building. A comfortable environment, however, is broader than just temperature and humidity. Comfort requirements that are typically impacted by the HVAC system include: □ Dry-bulb temperature (Temperature of air measured by a thermometer freely hanged) □ Humidity □ Air movement □ Fresh air □ Cleanliness of the air □ Noise levels			
II. 7	Two energy saving measures in cooling tower *Any two-3 marks*	3	3	
	Turn off unnecessary cooling tower fans when loads are diminished. Cover hot water basins (to minimize algae growth that contributes to fouling). Balance flow to cooling tower hot water basins. Periodically clean clogged cooling tower water distribution nozzles.			

II. 8	**Role of ESCO** *3 marks*	3	3	
				
II. 9	**Cash Flow model** *explanation-3 marks*	3	3	
	Cash Flow (CF) is the increase or decrease in the amount of money a business, institution, or individual has. In finance, the term is used to describe the amount of cash (currency) that is generated or consumed in a given time period. Cash flow calculations provide information on profitability, quality of earnings, liquidity, risks, capital requirements, future growth, dividends, etc. They are some of the most important tools for value investment analysis of investment opportunities.Cash flows are classified as operating, investing, or financing activities on the statement of cash flows, depending on the nature of the transaction.			

II. 10	**Average rate of return method** *Any three Merits:1 marks each*	3	3	
	• It is very simple to understand and use. • It can readily be calculated by using the accounting data. • This method takes into account saving over the entire economic life of the project. Therefore, it provides a better means of comparison of projects than the payback period. • This method through the concept of "net earnings" ensures a compensation of expected profitability of the projects.			
	PART C			42
III	**Ultrasonic flow meters and Stroboscopic Non contact tachometers** *3.5 marks each.*	3.5*2	7	
	• Ultrasonic flow meters are used to estimate fluid flow without having to penetrate piping.Ultrasonic flow meters operate based on one of two methods. Some use the the frequency shift (i.e. Doppler Effect) experienced by an ultrasonic signal as it is reflected by bubbles or particles (i.e., discontinuities) entrained by a flowing fluid.The magnitude of the frequency shift is indicative of the velocity of the fluid. Other ultrasonic flow meters are able to estimate the velocity of a clear (i.e., free of entrained particles or bubbles) liquid.*Application:* Ultrasonic flow meters can be used to estimate flow rates entering or leaving a pump. For example, the instrument can be used to ensure that flow rates are maintained as efficiency improvements (i.e., reducing motor size, and re-plumbing to reducing frictional losses) are made to plumbing systems			

	A tachometer is an instrument used to measure the rotational speed of a shaft or wheel in revolutions per minute (rpm). A stroboscopic tachometer employs a variable-frequency, flashing light which makes the rotating component appear to stand still when the frequencies match. This allows the users to measure the rotational speed without contacting the object in question *Application:* Stroboscopes are typically used to determine the mechanical loading of motors. By measuring a motor speed in rpm and electrical consumption, its efficiency can be determined. Non-contact stroboscopes are also commonly used to measure fan speeds and determine fan output (using the design fan curve).			
IV	**Duties of Energy managers** *7 marks*	7	7	

	• Report to BEE and State level Designated Agency once a year the information with regard to the energy consumed and action taken on the recommendation of the accredited energy auditor, as per BEE Format. • Establish an improved data recording, collection and analysis system to keep track of energy consumption. • Provide support to Accredited Energy Audit Firm retained by the company for the conduct of energy audit • Provide information to BEE as demanded in the Act, and with respect to the tasks given by a mandate, and the job description. • Prepare a scheme for efficient use of energy and its conservation and implement such scheme keeping in view of the economic stability of the investment in such form and manner as may be provided in the regulations of the Energy Conservation Act.			
V	**Detailed energy audit steps** *7 marks*	7	7	

Ten Steps Methodology for Detailed Energy Audit

Step No	PLAN OF ACTION	PURPOSE / RESULTS
	Phase I –Pre Audit Phase	
Step 1	▪ Plan and organise ▪ Walk through Audit ▪ Informal Interview with Energy Manager, Production / Plant Manager	▪ Resource planning, Establish/organize a Energy audit team ▪ Organize Instruments & time frame ▪ Macro Data collection (suitable to type of industry.) ▪ Familiarization of process/plant activities ▪ First hand observation & Assessment of current level operation and practices
Step 2	▪ Conduct of brief meeting / awareness programme with all divisional heads and persons concerned (2-3 hrs.)	▪ Building up cooperation ▪ Issue questionnaire for each department ▪ Orientation, awareness creation
	Phase II –Audit Phase	
Step 3	▪ Primary data gathering, Process Flow Diagram, & Energy Utility Diagram	▪ Historic data analysis, Baseline data collection ▪ Prepare process flow charts ▪ All service utilities system diagram (Example: Single line power distribution diagram, water, compressed air & steam distribution. ▪ Design, operating data and schedule of operation ▪ Annual Energy Bill and energy consumption pattern (Refer manual, log sheet, name plate, interview)
Step 4	▪ Conduct survey and monitoring	▪ Measurements : Motor survey, Insulation, and Lighting survey with portable instruments for collection of more and accurate data. Confirm and compare operating data with design data.
Step 5	▪ Conduct of detailed trials /experiments for selected energy guzzlers	▪ Trials/Experiments: - 24 hours power monitoring (MD, PF, kWh etc.). - Load variations trends in pumps, fan
Step6	▪ Analysis of energy use	▪ Energy and Material balance & energy loss/waste analysis
Step 7	▪ Identification and development of Energy Conservation (ENCON) opportunities	▪ Identification & Consolidation ENCON measures □ Conceive, develop, and refine ideas □ Review the previous ideas suggested by unit personal □ Review the previous ideas suggested by energy audit if any □ Use brainstorming and value analysis techniques □ Contact vendors for new/efficient technology
Step 8	▪ Cost benefit analysis	▪ Assess technical feasibility, economic viability and prioritization of ENCON options for implementation ▪ Select the most promising projects ▪ Prioritise by low, medium, long term measures
Step9	▪ Reporting & Presentation to the Top Management	▪ Documentation, Report Presentation to the top Management.
	Phase III –Post Audit phase	
Step10	▪ Implementation and Follow-up	Assist and Implement ENCON recommendation measures and Monitor the performance □ Action plan, Schedule for implementation □ Follow-up and periodic review

VI	**Benchmarking** *7 marks*	7	7	
	• Benchmarking of energy consumption internally (historical / trend analysis) and externally (across similar industries) are two powerful tools for performance assessment and logical evolution of avenues for improvement. Historical data well documented helps to bring out energy consumption and cost trends month-wise / day-wise.External benchmarking relates to inter-unit comparison across a group of similar units. Benchmarking energy performance permits • Quantification of fixed and variable energy consumption trends vis-à-vis production levels • Comparison of the industry energy performance with respect to various production levels (capacity utilization) • Identification of best practices (based on the external benchmarking data) • Scope and margin available for energy consumption and cost reduction			
VII	**Types and classification of Furnaces** *7 mark*	7	7	

Based on the method of generating heat, furnaces are broadly classified into two types namely combustion type (using fuels) and electric type. In the case of a combustion type furnace, depending upon the kind of combustion, it can be broadly classified as oil fired, coal fired or gas fired.

- Based on the mode of charging of material, furnaces can be classified as (i) Intermittent or Batch type furnaces or Periodical furnaces and (ii) Continuous furnaces.
- Based on mode of waste heat recovery as recuperative and regenerative furnaces.
- Another type of furnace classification is made based on mode of heat transfer, mode of charging and mode of heat recovery as shown in the Figure 4.1 below.

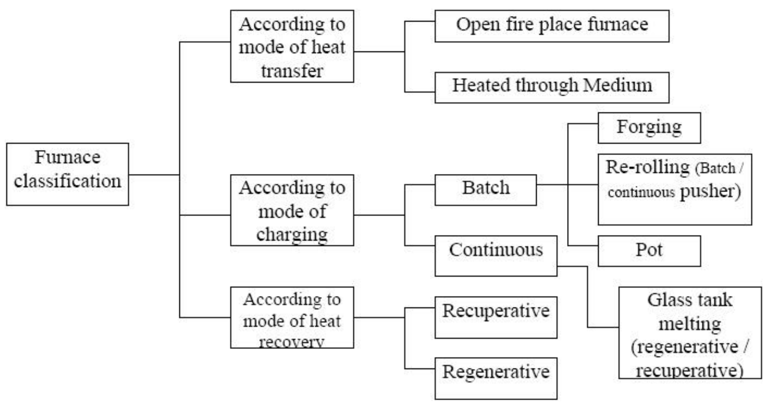


Figure 4.1 : Furnace Classification

VIII	**Energy conservation opportunities in Boilers** *each point 1 mark*	7	7	

1. Stack Temperature

The stack temperature (Temperature used to remove water vapor in the exhaust condenses on the stack walls) should be as low as possible. An estimated 1% efficiency loss occurs with every 22 °C increase in stack temperature

2. Feed Water Preheating using Economizer

Feed Water Preheating using economizer would reduce the exit temperature to 65 °C thereby increasing thermal efficiency by 5%.

3. Combustion Air Preheat

Combustion air preheating is an alternative to feed water preheating. In order to improve thermal efficiency by 1%, the combustion air temperature must be raised by 20 °C using pre- heater

4 Incomplete Combustion

Incomplete combustion can arise from a shortage of air or surplus of fuel or poor distribution of fuel. It is usually obvious from the color or smoke, and must be corrected immediately

5. Excess Air Control

Controlling excess air to an optimum level always results in reduction in flue gas losses; for every 1% reduction in excess air there is approximately 0.6% rise in efficiency.

6. Radiation and Convection Heat Loss

Repairing or augmenting insulation can reduce heat loss through boiler walls and piping.

7. Automatic Blowdown Control

<table>
<tr><td>IX</td><td>Classification of insulation
7 marks</td><td>3.5+3
.5</td><td>7</td><td></td></tr>
<tr><td></td><td colspan="2">The Insulation can be classified into three groups according to the temperature ranges for which they are used.

Low Temperature Insulations (up to 90°C)

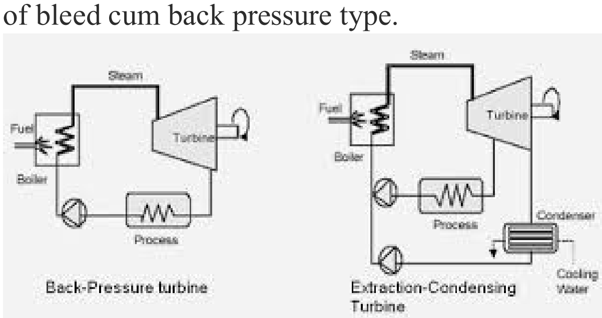
This range covers insulating materials for refrigerators, cold and hot water systems, storage tanks, etc. The commonly used materials are Cork, Wood, 85% magnesia, Mineral Fibers, Polyurethane and expanded Polystyrene, etc

Medium Temperature Insulations (90 – 325°C)

Insulators in this range are used in low temperature, heating and steam raising equipment, steam lines, flue ducts etc. The types of materials used in this temperatures range include 85% Magnesia, Asbestos, Calcium Silicate and Mineral Fibers etc.

High Temperature Insulations (325° C – above)

Typical uses of such materials are super heated steam system, oven dryer and furnaces etc. The most extensively used materials in this range are Asbestos, Calcium Silicate, Mineral Fibre, Mica and Vermiculite based insulation, Fireclay or Silica based insulation and Ceramic Fibre.</td><td></td><td></td></tr>
<tr><td>X</td><td>Back pressure steam turbine cogeneration system
figure-3.5marks,explanation-3.5 marks</td><td>7</td><td>7</td><td></td></tr>
</table>

	● This system is employed where the end-user needs thermal energy at two different temperature levels. ● Back pressure refers to steam turbines that exhaust steam at above atmospheric pressures used for drive application (Sugar mill drives, Sugar fibrizor / Shredder drive, Pump drives). ● The power generation is incidental to the process of steam demand. ● The back pressure turbine with one controlled extraction point is possible, this extraction steam is also used to meet process steam demand at intermediate pressure when volume demand is high and pressure variations cannot be tolerated. ● These turbines are of extraction cum back pressure type. The back pressure turbine may also have bleed points (uncontrolled extractions) to satisfy steam demands at intermediate pressures. This provision is applicable when the bleed (medium pressure) steam volume demand is low and pressure variations can be tolerated. These turbines are of bleed cum back pressure type.  			
XI	**Seven energy conservation opportunities in Lighting system** *Any seven-1 marks each*	7	7	

	1. Installation of compact fluorescent lamps (CFLs) in place of incandescent lamps. 2. Installation of energy-efficient fluorescent lamps in place of “conventional” fluorescent lamps. 3. Installation of occupancy/motion sensors to turn lights on and off where appropriate. 4. Use an automated device, such as a key tag system, to regulate the electric power in a room. 5.Offer nightlights to prevent the bathroom lights from being left on all night. 6. Replace all exit signs with light emitting diode (led) exit signs. 7. Use high efficiency (hid) exterior lighting 8.Add lighting controls such as photo sensors or time clocks			
XII	E**nergy management opportunities in Electric motors** *Any seven points,each carries1 marks*	7	7	
	Energy Saving Opportunities ● Improve controls ● Improve Power factor ● Improve cooling ● Turn motors off if not in use ● Schedule Regular maintenance ● Provide Regular lubrication Reduce Peak Demand by rescheduling motor operation			
XIII	**Payback period calculation** *7 marks*	7	7	

	Star rated device	Non Star rated device
cost	Rs 34000/-	Rs 20000/-
Star level	5 star	No star
Annual Electricity consumption	816 units	2080 units

Annual energy savings=2080-816=1264 units

Annual savings in rupees @ rupees 7 =1264*7=8848

payback period=34000/8848 =3.84 =4 years

	XIV	**Payback period calculation** *7 marks*	3.5 + 3.5	7	
		Annual energy savings = 486000-42000 = 444000 Payback period = 2220000/444000 = 5years			

Scoring Indicators

Model Question Paper II

Energy Management and Auditing

Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			9
I. 1	Role of energy manager *definition: 1 mark*	1	1	
	The role of an Energy Manager (EM) involves facilitating energy conservation by identifying and implementing various options for saving energy, leading awareness programs, and monitoring energy consumption.			
I. 2	**Energy management** *definition: 1 mark*	1	1	
	Energy management is the process of tracking and optimizing energy consumption to conserve usage in a building.			
I. 3	**Two Parts of boilers** *0.5 marks each mark*	0.5+0.5	1	
	Burner – The burner is the component of the boiler that provides the heat that heats the water of the system. ... Heat exchanger – The heat exchanger of a boiler allows the heat from the burner to heat the water in the system.			

I. 4	**Cogeneration** *definition: 1 mark*	1	1	
	Cogeneration or Combined Heat and Power (CHP) is defined as the sequential generation of two different forms of useful energy from a single primary energy source, typically mechanical energy and thermal energy.			
I. 5	**Two commercial waste heat recovery devices** *Any two-0.5 marks each*	0.5+0.5	1	
	a) Recuperator b) Radiation/Convective Hybrid Recuperator c) Ceramic Recuperator d) Regenerator e) Heat Wheels f) Heat Pipe g) Economiser h) Thermocompressor i) Direct Contact Heat Exchanger			
I. 6	**Type of Lamp which is an energy efficient replacement for Fluorescent Lamp** *1 mark*	1	1	
	LED			
I. 7	**Coefficient Of Performance** *definition: 1 mark*	1	1	
	The Coefficient Of Performance (COP) is a performance rating that tells us how effective a heat pump or air conditioner is at transferring heat versus the amount of electrical power it consumes.			
I. 8	**Advantage of IRR** *Any two-0.5 marks each*	0.5+0.5	1	

	• It incorporates the time value of money into the calculation. • It is a simple calculation. • It offers a method to rank projects for profitability. . • It works well with other evaluation factors. • It is not linked with the required rate of return.			
I. 9	**Advantage of simple payback period** *Any two-0.5 marks each*	0.5+0.5	1	
	• A longer payback period indicates capital is tied up. • Focus on early payback can enhance liquidity. • Investment risk can be assessed through the payback method. • Shorter term forecasts. • This is a more reliable technique.			
	PART B			24
II. 1	**Responsibilities of Energy manager** *Any six points-0.5 marks each*	0.5*6	3	

	• Prepare an annual activity plan and present to management concerning financially attractive investments to reduce energy costs • Establish an energy conservation cell within the firm with management's consent about the mandate and task of the cell. • Initiate activities to improve monitoring and process control to reduce energy costs. • Analyze equipment performance with respect to energy efficiency • Ensure proper functioning and calibration of instrumentation required to assess level of energy consumption directly or indirectly. • Prepare information material and conduct internal workshops about the topic for other staff. • Improve disaggregating of energy consumption data down to shop level or profit center of a firm. • Establish a methodology on how to accurately calculate the specific energy consumption of various products/services or activity of the firm. •Develop and manage training programmes for energy efficiency at operating levels. • Coordinate nomination of management personnel to external programs.			
II. 2	**Principle of combustion of fuels**	3	3	

	Principle of Combustion **Combustion** refers to the rapid oxidation of fuel accompanied by the production of heat, or heat and light. Complete combustion of a fuel is possible only in the presence of an adequate supply of oxygen. Oxygen (O_2) is one of the most common elements on earth making up 20.9% of our air. Rapid fuel oxidation results in large amounts of heat. Solid or liquid fuels must be changed to a gas before they will burn. Usually heat is required to change liquids or solids into gases. Fuel gases will burn in their normal state if enough air is present. Most of the 79% of air (that is not oxygen) is nitrogen, with traces of other elements. Nitrogen is considered to be a temperature reducing dilutant that must be present to obtain the oxygen required for combustion. Nitrogen reduces combustion efficiency by absorbing heat from the combustion of fuels and diluting the flue gases. This reduces the heat available for transfer through the heat exchange surfaces. It also increases the volume of combustion by-products, which then have to travel through the heat exchanger and up the stack faster to allow the introduction of additional fuel air mixture. This nitrogen also can combine with oxygen (particularly at high flame temperatures) to produce oxides of nitrogen (NO_x), which are toxic pollutants.		•	
II. 3	**List any three Prime Movers for Cogeneration and give a brief outline about each** *Any 3(3x.5) + Explanation each (3x.5)*	3	3	

	• Steam turbines. • Gas turbines. • Reciprocating engines. **Steam Turbine** Steam turbines come in a huge range of sizes, with the smallest being 80kW and the largest being 500,000kW. **Gas Turbine** Gas turbines are another prime mover choice, coming in sizes from 1,000 to 30,000kW. **Spark Ignition Reciprocating Engine** Spark ignition reciprocating engines come in sizes ranging from 50 to 4,000kW and are a quite common prime mover choice.			
II. 4	**Boiler system** *explanation-3 marks*	3	3	

	. • The boiler system comprises: feed water system, steam system and fuel system. • The feed water system provides water to the boiler and regulates it automatically to meet the steam demand. Various valves provide access for maintenance and repair.The steam system collects and controls the steam produced in the boiler• Steam is directed through a piping system to the point of use. Throughout the system, steam pressure is regulated using valve and checked with steam pressure gauges.The fuel system includes all equipment used to provide fuel to generate the necessary heat.The equipment required in the fuel system depends on the type of fuel used in the system.The water supplied to the boiler that is converted into steam is called feed water.			
II. 5	**Combustion controls various types** *1 marks each*	1*3	3	

	ON/OFF Control The simplest control, ON/OFF control means that either the burner is firing at full rate or it is OFF. This type of control is limited to small boilers. **High/Low/OFF Control** Slightly more complex is HIGH/LOW/OFF system where the burner has two firing rates. The burner operates at slower firing rate and then switches to full firing as needed. Burner can also revert to low firing position at reduced load. This control is fitted to medium sized boilers. **Modulating Control** The modulating control operates on the principle of matching the steam pressure demand by altering the firing rate over the entire operating range of the boiler. Modulating motors use conventional mechanical linkage or electric valves to regulate the primary air, secondary air, and fuel supplied to the burner. Full modulation means that boiler keeps firing, and fuel and air are carefully matched over the whole firing range to maximize thermal efficiency.			
II. 6	**Necessity of cogeneration** *Any 3 points 1 marks each*	3	3	

	• Cogeneration or Combined Heat and Power (CHP) is defined as the sequential generation of two different forms of useful energy from a single primary energy source, typically mechanical energy and thermal energy. • Mechanical energy may be used either to drive an alternator for producing electricity, or rotating equipment such as motor, compressor, pump or fan for delivering various services. • Thermal energy can be used either for direct process applications or for indirectly producing steam, hot water, hot air for dryers or chilled water for the process cooling. • Cogeneration provides a wide range of technologies for application in various domains of economic activities. The overall efficiency of energy use in cogeneration mode can be up to 85 per cent and above in some cases. Along with the saving of fossil fuels, cogeneration also allows to reduce the emission of greenhouse gasses			
II. 7	**Energy saving measures in Air Conditioning System** *Any three-1 marks each*	3	3	

	1) Ensure air intake to the compressor is not warm and humid by locating compressors in a well ventilated area or by drawing cold air from outside. Every 4°C rise in air inlet temperature will increase power consumption by 1 percent. 2)Clean air-inlet filters regularly. Compressor efficiency will be reduced by 2 percent for every 250 mm WC pressure drop across the filter. 3) Keep compressor valves in good condition by removing and inspecting them once every six months. Worn-out valves can reduce compressor efficiency by as much as 50 percent. 4) Install manometers across the filter and monitor the pressure drop as a guide to replacement of elements.			
II. 8	**ESCO concept** *explanation-3 marks*	3	3	

	ESCOs are usually companies that provide a complete energy project service, from assessment to design to construction or installation, along with engineering and project management services, and financing. In one way or another, the contract involves the capitalization of all of the services and goods purchased, and repayment out of the energy savings that result from the project. In performance contracting, an end-user (such as an industry, institution, or utility) seeking to improve its energy efficiency, contracts with ESCO for energy efficiency services and financing.In some contracts, the ESCOs provide a guarantee for the savings that will be realized, and absorbs the cost if real savings fall short of this level. Typically, there will be a risk management cost involved in the contract in these situations. Insurance is sometimes attached, at a cost, to protect the ESCO in the event of a savings shortfall. Energy efficiency projects generate incremental cost savings as opposed to incremental revenues from the sale of outputs. The energy cost savings can be turned into incremental cash flows to the lender or ESCO based on the commitment of the energy user (and in some cases, a utility) to pay for the savings .			
II. 9	**Benefits of Energy monitoring and Targeting** *Any six points-0.5 marks each*	0.5*6	3	

	✓ Identify and explain an increase or decrease in energy use ✓ Draw energy consumption trends (weekly, seasonal, operational) ✓ Improve energy budgeting corresponding to production plans ✓ Observe how the organization reacted to changes in the past ✓ Determine future energy use when planning changes in operations ✓ Diagnose specific areas of wasted energy ✓ Develop performance targets for energy management programs / energy action plans ✓ Manage energy consumption rather than accept it as a fixed cost that cannot be controlled.			
II. 10	**Need and benefits of star labeling** *need-1.5 mark's,benefits-1.5 marks*	1.5*2	3	
	. A higher star rating means a more efficient appliance, which means a lower electricity bill every month. Most modern households have multiple appliances that consume a lot of electricity such as air conditioners, refrigerators, and even ceiling fans. Benefits of Star Labeling Star rating schemes allow us to compare the operating cost and environmental performance of similar products. This allows us to make better choices and see the potential ongoing future costs and greenhouse gas impacts our decisions may have			

	PART C			42
III	**Energy conservation and its importance** *definition -2 marks,importance-5 marks.*	3.5*2	7	

	• Energy conservation is the effort made to reduce the consumption of energy by using less of an energy service. This can be achieved either by using energy more efficiently (using less energy for a constant service) or by reducing the amount of service used .Most of our energy use comes from fossil fuels like petroleum and coal that provide electricity and gas to power our growing energy needs. These resources are non-renewable which means that we will eventually run out. Conserving energy not only helps to conserve resources but also translates into financial savings. ● We use energy faster than it can be produced. Coal, oil and natural gas - the most utilized sources take thousands of years for formation. ●Energy resources are limited. India has approximately 1% of the world's energy resources but it has 16% of the world population. ●Most of the energy resources we use cannot be reused and renewed. Non - renewable energy resources constitute 80% of the fuel use. It is said that our energy resources may last only for another 40 years or so.			

	• Save energy to reduce pollution. Energy production and use account to large proportion of air pollution and more than 83 percent of greenhouse gas emissions • Energy saved is energy generated. When we save one unit of energy, it is equivalent to 2 units of energy produced.			
IV	**Radiation Pyrometers and Power Analyzers** *3.5 marks each*	3.5*2	7	
	Radiation Pyrometers These thermometers rely on the electromagnetic radiation emitted by solids or fluids. The radiation is characteristic of their temperature. A lens focuses the infrared energy on the active detecting surface. The heart of the infrared thermometer is the detecting surface, which absorbs infrared energy and converts it to an electrical voltage or current. The accuracy of temperature measurements by infrared instruments depends on the absorption, reflection, and transmission characteristics of the radiative flux *Application:* Non-contact infrared thermometers, also known as heat guns, are very useful for measuring surface temperatures of steam lines, boiler surfaces, processes temperatures, etc. The primary use of infrared sensors in an energy management program is to detect building or equipment thermal losses, pinpoint insulation or weatherization needs, identify electrical hot spots, and locate unseen motor friction points			

	Power Analyzer A power analyzer is used to measure the flow of power (w) in an electrical system. This refers to the rate of electrical transferral between a power source and a sink, hence the alternative expression of power as energy per second (J/s). Power analyzers are used for measuring a host of aspects of electrical power for applications which include testing power electronics, inverters, motors and drives, lighting, home appliances, office equipment, power supplies, industrial machinery and more.			
V	**Targeted energy audit** *explanation-7 marks*	7	7	
	Targeted Energy Auditing: ● Based on the outcome of the preliminary audit result.It Provides data and analysis on specified target projects. ● An organization may target its lighting system or boiler system or compressed air system with a view to bring out energy savings. ● Targeted audits therefore involve detailed surveys of the target subjects/areas with analysis of energy flows and costs associated with those targets.			
VI	**Preliminary Energy Audit Methodology** *explanation-7 marks*	7	7	

	• Establish energy consumption in the organization • Estimate the scope for saving • Identify the most likely (and the easiest areas for attention • Identify immediate (especially no-/low-cost) improvements/ savings • Set a 'reference point' • Identify areas for more detailed study/measurement • Preliminary energy audit uses existing, or easily obtained data.			
VII	**Energy conservation opportunities in Furnaces** *Any seven points1 mark each*	7	7	

	1) Complete combustion with minimum excess air 2) Correct heat distribution 3) Operate furnace at the desired temperature 4) Reduce heat losses from furnace openings 5) Maintain correct amount of furnace draught 6) Optimum capacity utilization of furnace will give maximum thermal efficiency 7) Waste heat recovery from the flue gasses improves system efficiency 8) Minimum refractory losses The appropriate choice of refractory and insulation materials goes a long way in achieving fairly high fuel savings in industrial furnaces 9) Use of Ceramic Coatings in the furnace chamber promotes rapid and efficient transfer of heat, thereby extending the life of refractories.			
VIII	**classification of Refractories** *classification 7 marks*	7	7	

	Table 5.3 Classification Based on Chemical Composition			
IX	**Energy conservation opportunities in Compressed Air system** *Any seven points1 mark each*	7	7	

Table 5.3 Classification Based on Chemical Composition

Classification Based on Chemical Composition	Examples
ACID--which readily combines with bases.	Silica, Semisilica, Aluminosilicate.
BASIC--which consists mainly of metallic oxides which resist the action of bases.	Magnesite, chromemagnesite, Dolomite.
NEUTRAL--which doesn't combine; neither with acids nor bases.	Chrome, Pure. Alumina
Special	Carbon, Silicon Carbide, Zirconia.
Classification based on end use	Blast furnace Casting Pit
Classification based on method of manufacture	- Dry Press Process - Fused Cast - Hand Moulded - Formed Normal, fired or Chemically bonded.) - Unformed (Monolithics-plastics, Ramming Mass, Gunning Castable, Spraying.)

	• Ensure air intake to the compressor is not warm and humid by locating compressors in a well ventilated area or by drawing cold air from outside. Every 4°C rise in air inlet temperature will increase power consumption by 1 percent. • Clean air-inlet filters regularly. Compressor efficiency will be reduced by 2 percent for every 250 mm WC pressure drop across the filter. • Keep compressor valves in good condition by removing and inspecting once every six months. Worn-out valves can reduce compressor efficiency by as much as 50 percent. • Install manometers across the filter and monitor the pressure drop as a guide to replacement of elements. • Compressor free air delivery test (FAD) must be done periodically to check the present operating capacity against its design capacity and corrective steps must be taken if required. • Fouled inter-coolers reduce compressor efficiency and cause more water condensation in air receivers and distribution lines resulting in increased corrosion. Periodic cleaning of intercoolers must be ensured. • Reduce compressor delivery pressure, wherever possible, to save energy.			
X	**Energy conservation opportunities in Refrigeration Plants** *Any seven points1 mark each*	7	7	

	a) Cold Insulation Insulate all cold lines / vessels using economic insulation thickness to minimize heat gains; and choose appropriate (correct) insulation. b) Building Envelope Optimize air conditioning volumes by measures such as use of false ceiling and segregation of critical areas for air conditioning by air curtains. c) Building Heat Loads Minimization Minimize the air conditioning loads by measures such as roof cooling, roof painting, efficient lighting, pre-cooling of fresh air by air- to-air heat exchangers, variable volume air system, optimal Thermostat setting of temperature of air conditioned spaces, sun film applications, etc. d) Process Heat Loads Minimization Minimize process heat loads in terms of TR capacity as well as refrigeration level. e) At the Refrigeration A/C Plant Area i) Ensure regular maintenance of all A/C plant components as per manufacturer guidelines. ii) Ensure adequate quantity of chilled water and cooling water flows, avoid bypass flows by closing valves of idle equipment. iii) Minimize part load operations by matching loads and plant capacity on line; adopt variable speed drives for varying process load.			

XI	**Energy conservation opportunities in DG set** *Any seven points each carries1 marks*	7	7	
	a) Ensure fuel oil storage, handling and preparation as per manufacturers' guidelines/oil company data. b) Consider fuel oil additives in case they benefit fuel oil properties for DG set usage. c) Calibrate fuel injection pumps frequently. d) Ensure compliance with maintenance checklist. e) Ensure steady load conditions, avoiding fluctuations, imbalance in phases, harmonic loads. f) Carryout regular field trials to monitor DG set performance, and maintenance planning as per requirements. g) Ensure steady load conditions on the DG set, and provide cold, dust free air at intake			
XII	**Energy saving measures in electrolytic process** *each carries1 marks*	7	7	
	• Insulate plating tanks • Provide proper maintenance of electrodes and rectifiers • Recover waste heat • Use more efficient rectifiers (semiconductor vs. mercury arc) • Use more efficient controls ● Develop improved electrode design and materials to increase efficiency			

XIII	**Payback period calculation** *7 marks*	7	7	

	Star rated device	Non Star rated device
cost	Rs 32000/-	Rs 18000/-
Star level	5 star	No star
Annual Electricity consumption	800 units	2050 units

Annual energy savings=2050-800=1250 units

Annual savings in rupees @ rupees 8 =1250*8=10000

payback period=32000/10000 =3.2 =3 years

XIV	**Payback period calculation** *7 marks*	7	7	
	Simple payback period = Capital cost/Annual net savings Simple Payback period = 90/(23-5) = 5 years			